Week 3 – Seller & Inventory Management System

1. Seller & Inventory Table Design

The Seller table contains Seller ID, Seller Name, Email, Phone and Address. Seller ID is the Primary Key, while Email and Phone are UNIQUE and NOT NULL.

The Product table contains Product ID, Product Name, Price and Seller ID. Product ID is the Primary Key and Seller ID is a Foreign Key referencing the Seller table.

The Inventory table contains Inventory ID, Product ID, Seller ID, Stock Quantity, Stock Status and Last Updated. Inventory ID is the Primary Key. Product ID and Seller ID are Foreign Keys. Stock Quantity uses a CHECK constraint to prevent negative values.

2. Sample Seller & Inventory Data

Sample sellers used are Ravi Traders, Kumar Electronics and Siva Stores.

Sample products include Keyboard, Mouse, Monitor, USB Cable, Headphones and Webcam.

The inventory contains different stock quantities and stock statuses such as Available and Out of Stock.

4. Inventory Analysis Report

The inventory analysis identifies available products, out-of-stock products and products having stock below 10 units.

In the sample data, Monitor is out of stock. Mouse and Headphones have stock below 10 units. Webcam has the highest stock quantity with 30 units. The average inventory quantity is 13.83 units.

5. Query Output Screenshots

The following SQL query outputs should be captured from MySQL Workbench:

1. Seller-wise Product Report
2. Available Products
3. Out-of-Stock Products
4. Products with Stock Less Than 10
5. Updated Stock Quantity
6. Highest Stocked Product
7. Average Inventory Quantity